

Overview of Precision and recall

Subject: Precision and recall

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

1.

Probabilistic definition Precision and recall can be interpreted as (estimated) conditional probabilities: Precision is given by $P(C = P | \hat{C} = P)$ while recall is given by $P(\hat{C} = P | C = P)$, where $\hat{C}$ is the predicted class and C is the actual class (i.e. $C = P$ means the actual class is positive). Both quantities are, therefore, connected by Bayes' theorem.

2.

Quantitative finance. A model may output a continuous buy/sell signal proportional to its conviction for each asset at a given moment. Applying a fixed threshold (e.g. signal > 0.1 = buy) discards conviction magnitude and conflates strong and mild signals, limiting the model's ability to correctly distinguish a strong buy from a mild one. DNA methylation analysis. Sequencing tools detect differentially methylated regions whose underlying signal is continuous, typically ranging from -1 to +1. Forcing this into a binary framework introduces similar distortions by treating all positive deviations as equivalent regardless of magnitude.

3.

This measure is approximately the average of the two when they are close, and is more generally the harmonic mean, which, for the case of two numbers, coincides with the square of the geometric mean divided by the arithmetic mean. There are several reasons that the F-score can be criticized, in particular circumstances, due to its bias as an evaluation metric. This is also known as the F_1 measure, because recall and precision are evenly weighted. It is a special case of the general F_β measure (for non-negative real values of β):

4.

Precision vs. recall Both precision and recall may be useful in cases where there is imbalanced data. However, it may be valuable to prioritize one metric over the other in cases where the outcome of a false positive or false negative is costly. For example, in medical diagnosis, a false positive test can lead to unnecessary treatment and expenses. In this situation, it is useful to value precision over recall. In other cases, the cost of a false negative is high, and recall may be a more valuable metric. For instance, the cost of a false negative in fraud detection is high, as failing to detect a fraudulent transaction can result in significant financial loss.

5.

Definitions Let x_i denote the expected (ground-truth) signal and y_i the observed (predicted) signal at each data point i . Then, precision, recall, and F-score can be calculated as:

6.

Definition For classification tasks, the terms true positives, true negatives, false positives, and false negatives compare the results of the classifier under test with trusted external judgments. The terms positive and negative refer to the classifier's prediction (sometimes known as the expectation), and the terms true and false refer to whether that prediction corresponds to the external judgment (sometimes known as the observation). Let us define an experiment from P positive instances and N negative instances for some condition. The four outcomes can be formulated in a 2×2 contingency table or confusion matrix, as follows:

7.

In pattern recognition, information retrieval, object detection and classification (machine learning), precision and recall are performance metrics that apply to data retrieved from a collection, corpus or sample space. Precision (also called positive predictive value) is the fraction of relevant instances among the retrieved instances. Written as a formula:

8.

$$F_{\beta} = (1 + \beta^2) \cdot \text{precision} \cdot \text{recall} / (\beta^2 \cdot \text{precision} + \text{recall})$$

9.

$$P(C = P | C = P) = P(C = P, C = P) / P(C = P) = P(C = P) / P(C = P)$$

which is the probability for a positive classification.

10.

Both precision and recall are therefore based on relevance. Consider a computer program for recognizing dogs (the relevant element) in a digital photograph. Upon processing a picture which contains ten cats and twelve dogs, the program identifies eight dogs. Of the eight elements identified as dogs, only five actually are dogs (true positives), while the other three are cats (false positives). Seven dogs were missed (false negatives), and seven cats were correctly excluded (true negatives). The program's precision is then $5/8$ (true positives / selected elements) while its recall is $5/12$ (true positives / relevant elements). Adopting a hypothesis-testing approach, where in this case, the null hypothesis is that a given item is irrelevant (not a dog), absence of type I and type II errors (perfect specificity and sensitivity) corresponds respectively to perfect precision (no false positives) and perfect recall (no false negatives). More generally, recall is simply the complement of the type II error rate (i.e., one minus the type II error rate). Precision is related to the type I error rate, but in a slightly more complicated way, as it also depends upon the prior distribution of seeing a relevant vs. an irrelevant item. The above cat and dog example contained $8 - 5 = 3$ type I errors (false positives) out of 10 total cats (true negatives), for a

type I error rate of $3/10$, and $12 - 5 = 7$ type II errors (false negatives), for a type II error rate of $7/12$. Precision can be seen as a measure of quality, and recall as a measure of quantity. Higher precision means that an algorithm returns more relevant results than irrelevant ones, and high recall means that an algorithm returns most of the relevant results (whether or not irrelevant ones are also returned).